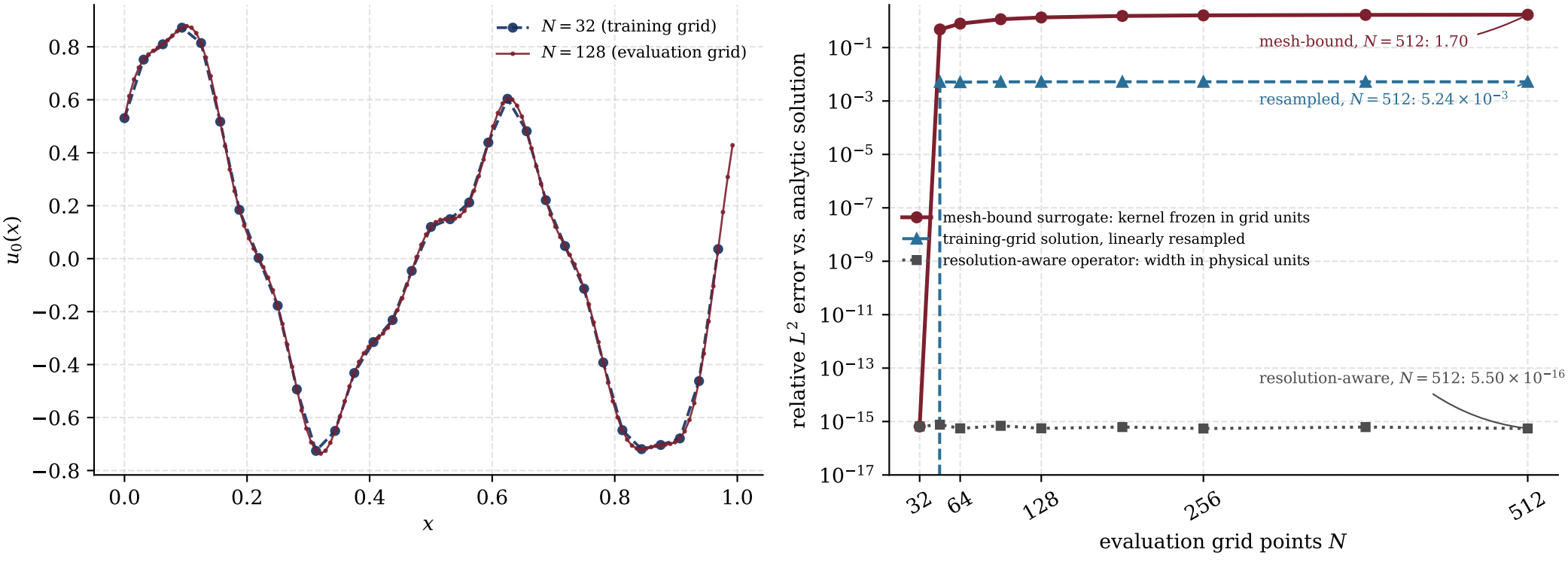
(a) One continuous field, two discretizations
$N = 32$ (training grid)
$N = 128$ (evaluation grid)
$u_0(x)$
x
(b) Error vs resolution, calibrated at $N = 32$
mesh-bound, $N = 512$: 1.70
resampled, $N = 512$: 5.24×10^{-3}
resolution-aware, $N = 512$: 5.50×10^{-16}
mesh-bound surrogate: kernel frozen in grid units
training-grid solution, linearly resampled
resolution-aware operator: width in physical units
relative L^2 error vs. analytic solution
evaluation grid points N